

CLIENT DELIVERY DOCUMENT

Hierarchia Manager Portal User Guide

Version: 1.0

Date: May 12, 2026

Prepared by: Shahroz Imran

Classification: Client Confidential

Purpose

Estimate the cost of using `google/gemini-3.1-pro-preview` and `anthropic/claude-sonnet-4-5` over **1,000 messages**.

Important Note

LLM pricing is **token-based**, not message-based. That means there is no single exact cost for 1,000 messages unless we also define the average:

- input tokens per message
- output tokens per message

This document uses current OpenRouter pricing and shows several realistic scenarios.

Pricing Used

Source checked on **2026-05-12** from OpenRouter model pages:

- **Gemini 3.1 Pro Preview**
 - Input: **\$2.00 / 1M tokens**
 - Output: **\$12.00 / 1M tokens**
 - Source: `https://openrouter.ai/google/gemini-3.1-pro-preview`
- **Claude Sonnet 4.5**
 - Input: **\$3.00 / 1M tokens**
 - Output: **\$15.00 / 1M tokens**
 - Source: `https://openrouter.ai/anthropic/claude-sonnet-4.5`

Formula

For **1,000 messages**:

```
Total cost = ((input_tokens_per_message × 1000) / 1,000,000 × input_rate)
              + ((output_tokens_per_message × 1000) / 1,000,000 × output_rate)
```

1,000-Message Scenarios

Scenario A: Light usage

Assumption per message:

- Input: tokens
- Output: tokens

Model	Input Cost	Output Cost	Total for 1,000 Messages
Gemini 3.1 Pro Preview	\$2.00	\$3.60	\$5.60
Claude Sonnet 4.5	\$3.00	\$4.50	\$7.50

Scenario B: Medium usage

Assumption per message:

- Input: tokens
- Output: tokens

Model	Input Cost	Output Cost	Total for 1,000 Messages
Gemini 3.1 Pro Preview	\$6.00	\$9.60	\$15.60
Claude Sonnet 4.5	\$9.00	\$12.00	\$21.00

Scenario C: Heavy usage

Assumption per message:

- Input: tokens
- Output: tokens

Model	Input Cost	Output Cost	Total for 1,000 Messages
Gemini 3.1 Pro Preview	\$16.00	\$24.00	\$40.00
Claude Sonnet 4.5	\$24.00	\$30.00	\$54.00

Simple Baseline Estimate

If you want one easy baseline for planning, use:

- Input: tokens per message
- Output: tokens per message

Model	Total for 1,000 Messages
Gemini 3.1 Pro Preview	\$10.00
Claude Sonnet 4.5	\$13.50

What This Means For Your Pipeline

Your current pipeline defaults are:

- **Validation model:** google/gemini-3.1-pro-preview
- **Vision model:** google/gemini-3.1-pro-preview
- **Summary model:** anthropic/claude-sonnet-4-5

That is a sensible cost/quality split because:

- **Gemini 3.1 Pro** is cheaper for repeated validation calls
- **Claude Sonnet 4.5** is more expensive, so keeping it limited to summaries helps control cost

Pipeline-Oriented Example

If we assume for **1,000 submissions**:

- 6 validation-rule calls per submission on Gemini
- 1 summary call per submission on Claude
- average rule call = 3,000 input / 400 output tokens
- average summary call = 2,000 input / 250 output tokens

Then the rough cost is:

Validation on Gemini

Per rule call:

- Input: $3,000 \times \$2 / 1\text{M} = \0.0060
- Output: $400 \times \$12 / 1\text{M} = \0.0048
- Total per rule call = **\$0.0108**

For 6,000 rule calls:

- **\$64.80**

Summaries on Claude

Per summary call:

- Input: $2,000 \times \$3 / 1M = \0.0060
- Output: $250 \times \$15 / 1M = \0.00375
- Total per summary call = **\$0.00975**

For 1,000 summary calls:

- **\$9.75**

Combined illustrative total

- **\$74.55 per 1,000 submissions**

Recommendation

If cost matters:

- keep **Gemini 3.1 Pro** for validation and vision
- keep **Claude Sonnet 4.5** only for summaries or high-value review steps

If you want a stricter budget estimate next, the best next step is to calculate from your real pipeline averages:

- average extracted text size
- average number of rules per submission
- average summary output length
- percentage of submissions using the vision path